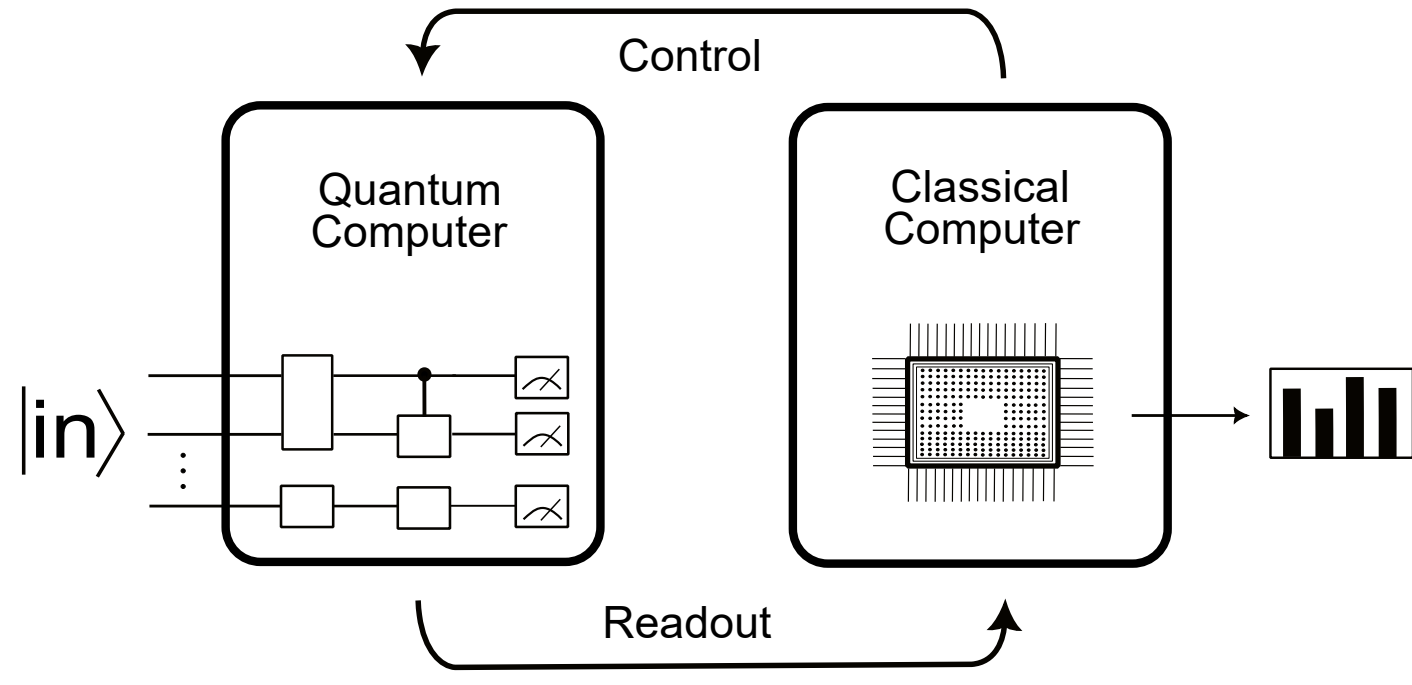


Motivation: Quantum and Classical Together

Open quantum systems exchange energy and coherence with their environment: their evolution is **non-unitary**. Quantum hardware executes only **unitary** gates – this work bridges that gap.



Hybrid loop: the CPU builds and controls the circuit, the QPU carries the exponentially large state, the measurement is read back.

- **Scaling:** 2^n amplitudes for n qubits
- **Task:** embed a non-unitary map into a larger unitary one
- **Tool:** Kraus operators + SVD + dilation
- **Test system:** the FMO light-harvesting complex

Qubits and the Bloch Sphere

A qubit may be in a **superposition** of the basis states:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Normalised, $|\alpha|^2 + |\beta|^2 = 1$, and $\text{Pr}(0) = |\alpha|^2$.

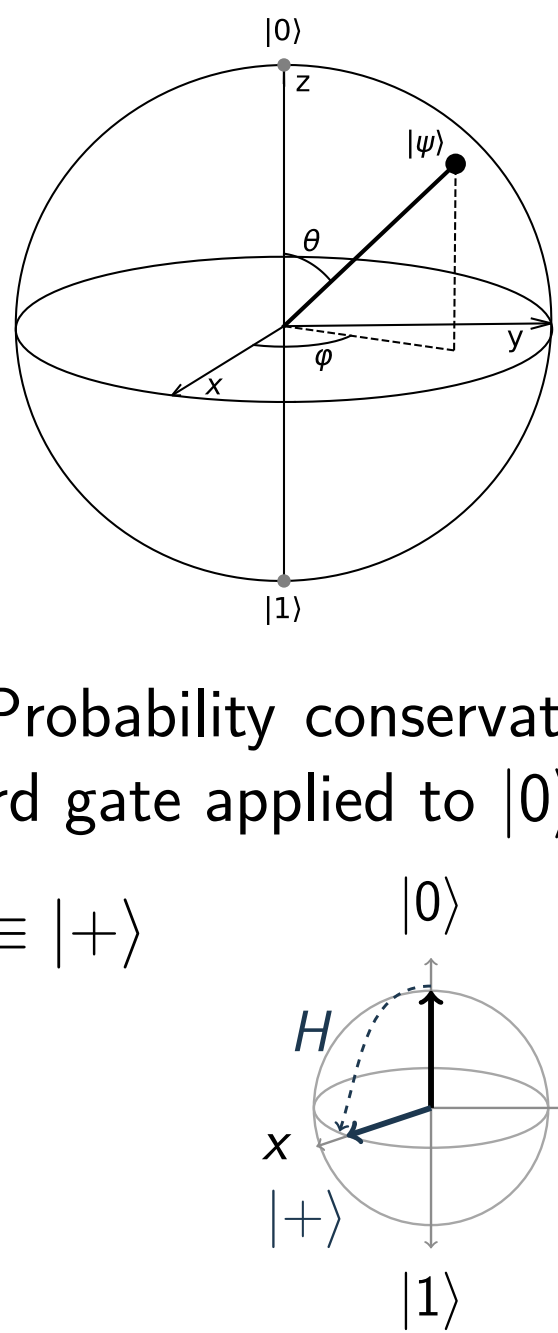
With $\alpha = \cos \frac{\theta}{2}$, $\beta = e^{i\varphi} \sin \frac{\theta}{2}$ the normalisation becomes two angles:

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\varphi} \sin \frac{\theta}{2} |1\rangle$$

A gate is a rotation of that arrow. Probability conservation demands $U^\dagger U = \mathbb{1}$. Example – the Hadamard gate applied to $|0\rangle$:

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \equiv |+\rangle$$

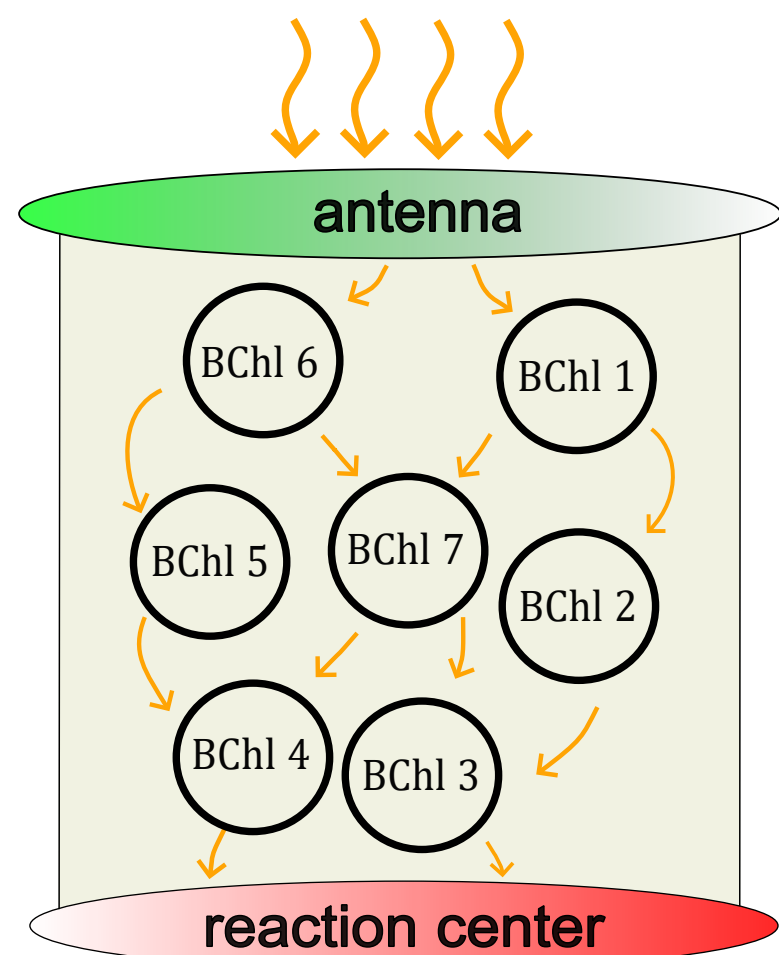
$\theta : 0 \rightarrow \frac{\pi}{2}$ at $\varphi = 0$: the arrow swings from the north pole onto the $+x$ axis – a definite $|0\rangle$ turns into an equal superposition. Applying H again brings it back, since $H|+\rangle = |0\rangle$.



The Fenna–Matthews–Olson Complex

Seven bacteriochlorophyll (BChl) molecules guide an excitation from the antenna to the reaction centre [1][3]. Absorption happens at ≈ 800 nm, in the infrared.

- BChl 1, 6 at the antenna; BChl 3, 4 at the reaction centre
- the excitation **delocalises** over all seven sites, mediated by a dipole–dipole interaction – no electron is ever exchanged



Each BChl is a **two-level system** (HOMO/LUMO). With at most one excitation in the complex this gives the site basis:

$$|i\rangle = |\phi_1^g\rangle \cdots |\phi_i^e\rangle \cdots |\phi_N^g\rangle$$

The Frenkel Exciton Model

Following Kreisbeck [2] the Hamiltonian splits into three parts:

$$\hat{H} = \hat{H}_S + \hat{H}_B + \hat{H}_{SB}$$

System – site energies and dipole–dipole coupling:

$$\hat{H}_S = \sum_m \epsilon_m^e |m\rangle\langle m| + \sum_{m \neq n} J_{mn} |m\rangle\langle n|$$

J_{mn} : excitation hops $m \rightarrow n \Rightarrow$ delocalisation. No electrons are exchanged.

Bath – nuclear vibrations as harmonic normal modes:

$$\mathcal{H}_B = \sum_{\xi} \left(\frac{p_{\xi}^2}{2m_{\xi}} + \frac{1}{2} m_{\xi} \omega_{\xi}^2 q_{\xi}^2 \right) = \sum_{\xi} \hbar \omega_{\xi} \left(b_{\xi}^{\dagger} b_{\xi} + \frac{1}{2} \right),$$

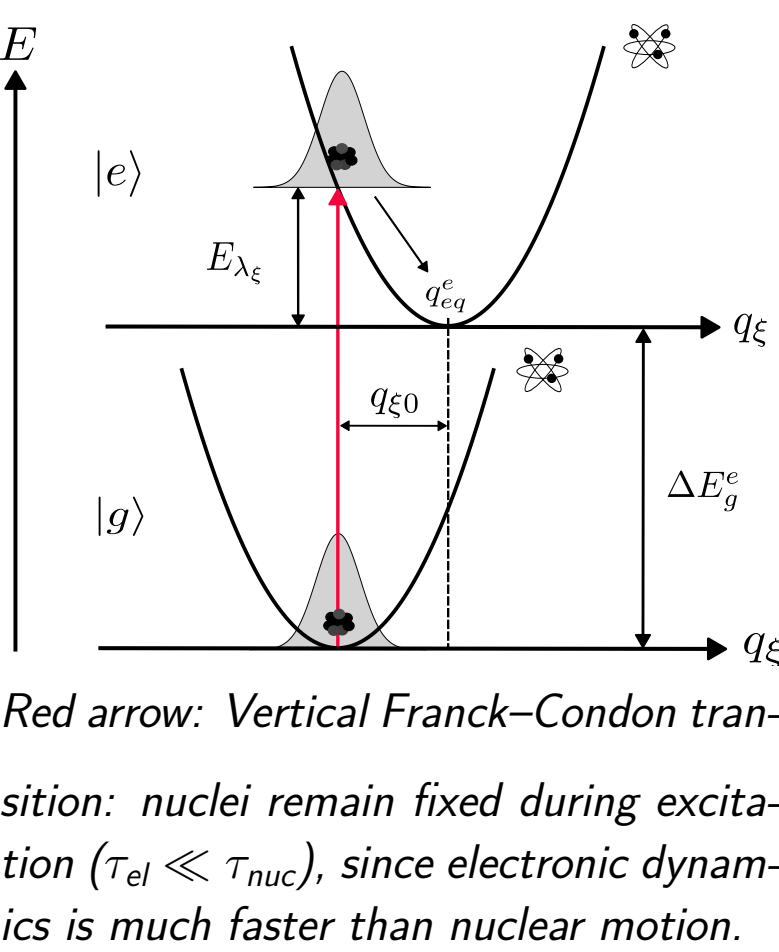
$b_{\xi}^{\dagger}$ creates a phonon $\hbar \omega_{\xi}$ in mode q_{ξ} .

Coupling. The nuclear dynamics follow potential energy surfaces (PES), harmonic around the equilibrium. In the excited state that equilibrium is shifted by $q_{\xi 0}$.

Expanding the shifted bath Hamiltonian leaves a term linear in $b_{\xi}^{\dagger} + b_{\xi}$ – that is the coupling, and it exists only while the site is excited. Relaxation into the new minimum dissipates the **reorganisation energy** $\lambda_m = \sum_{\xi} \hbar \omega_{\xi} m d_{\xi m}^2$.

$$\hat{H}_{SB} = \sum_m \sum_{\xi} \hbar \omega_{\xi} m d_{\xi m} (b_{\xi m}^{\dagger} + b_{\xi m}) \otimes |m\rangle\langle m|$$

Displacement $d_{\xi} = -\sqrt{m_{\xi} \omega_{\xi} / 2 \hbar} q_{\xi 0}$; every mode couples to every site.



Red arrow: Vertical Franck–Condon transition: nuclei remain fixed during excitation ($\tau_{el} \ll \tau_{nuc}$), since electronic dynamics is much faster than nuclear motion.

Markovian Dynamics: Lindblad $\rightarrow$ Kraus

Lindblad equation - the most general Markovian master equation:

$$\frac{d}{dt} \rho_S = \underbrace{-\frac{i}{\hbar} [H, \rho_S]}_{\text{unitary}} + \underbrace{\sum_{\alpha} \gamma_{\alpha} \left(L_{\alpha} \rho_S L_{\alpha}^{\dagger} - \frac{1}{2} \{ L_{\alpha}^{\dagger} L_{\alpha}, \rho_S \} \right)}_{\text{dissipator} \rightarrow \text{non-unitary}} := \mathcal{L} \rho_S$$

γ_{α} : transition rate $\rightarrow$ Fermi's golden rule. L_{α} : Lindblad operator $\rightarrow$ jump operator.

Kraus operators directly from it. For a small step dt the Ansatz $M_0 = \mathbb{1} + L_0 dt$, $M_{\alpha} = \sqrt{dt} L_{\alpha}$ and $\sum_k M_k^{\dagger} M_k = \mathbb{1}$ fix L_0 , which gives

$$M_0 = \mathbb{1} + \left(-\frac{i}{\hbar} H - \frac{1}{2} \sum_{\alpha} \gamma_{\alpha} L_{\alpha}^{\dagger} L_{\alpha} \right) dt, \quad M_{\alpha} = \sqrt{\gamma_{\alpha} dt} L_{\alpha}$$

Exact only to $\mathcal{O}(dt)$ – it forces small steps. The algorithm therefore takes the route below, which is exact for *any* t .

1. **Vectorise** $\Rightarrow$ linear, integrable in closed form:

$$\text{vec}(\rho_S(t)) = e^{\mathcal{L}t} \text{vec}(\rho_S(0)) \equiv \Lambda(t) \text{vec}(\rho_S(0))$$

$\mathcal{L}$ is time independent $\Rightarrow$ the step t may be **arbitrarily large**: no Trotterisation, no small-step error.

2. **Choi matrix** of the map, then diagonalise:

$$C(t) = \sum_{i,j} (E_{ij} \otimes \mathbb{1}) \Lambda(t) (\mathbb{1} \otimes E_{ij}) = \sum_k \sigma_k u_k u_k^{\dagger}$$

$\Lambda(t)$ is CPTP $\Rightarrow C(t)$ positive, all $\sigma_k \geq 0$.

3. **Kraus operators** read off the eigenvectors:

$$\text{vec}(M_k(t)) = \sqrt{\sigma_k(t)} u_k(t), \quad \rho_S(t) = \sum_k M_k \rho_S(0) M_k^{\dagger}$$

Why the Circuit Looks the Way It Does

The M_k are **not unitary**. An SVD isolates the non-unitary part in the diagonal matrix Σ :

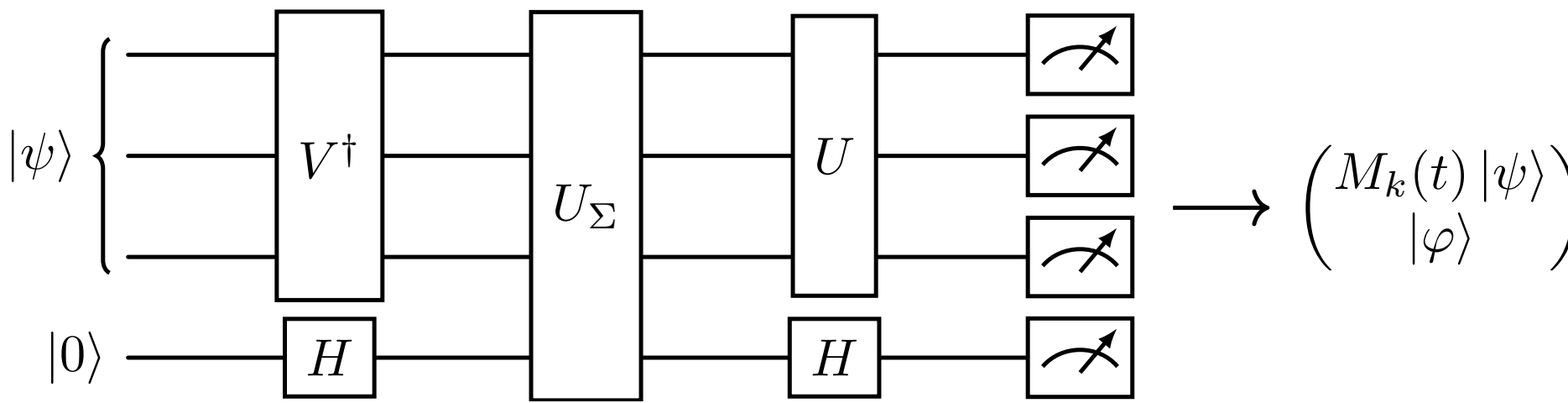
$$M_k(t) = U \Sigma V^{\dagger}$$

$U, V^{\dagger}$ are rotations and already unitary – only the rescaling Σ must be repaired.

Dilation into a unitary of twice the size – the price is exactly one ancilla qubit:

$$U_{\Sigma} = \begin{pmatrix} \Sigma_{+} & 0 \\ 0 & \Sigma_{-} \end{pmatrix}, \quad \Sigma_{\pm jj} = \sigma_{jj} \pm i \sqrt{1 - \sigma_{jj}^2}$$

$|\sigma_{jj}| \leq 1$ is guaranteed by complete positivity, so the root stays real.

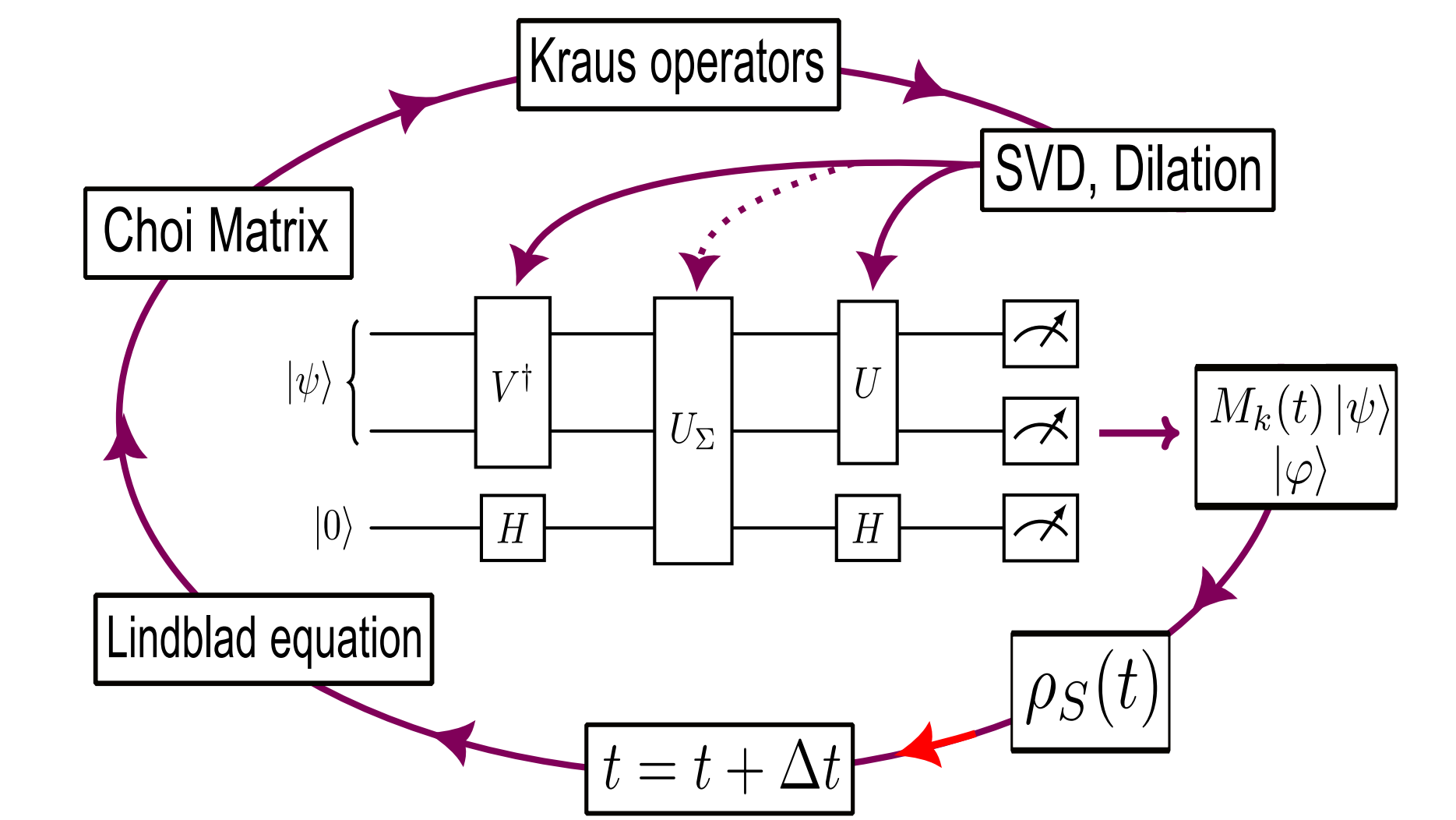


Seven-site FMO: three system qubits carry $|\psi\rangle$, the fourth is the ancilla, framed by two Hadamard gates.

Propagating $|\psi\rangle \otimes |0\rangle$ through the circuit and using $\Sigma_{+} + \Sigma_{-} = 2\Sigma$:

$$H U_{\Sigma} V^{\dagger} H |\psi\rangle \otimes |0\rangle = \frac{1}{2} \begin{pmatrix} U(\Sigma_{+} + \Sigma_{-}) V^{\dagger} |\psi\rangle \\ U(\Sigma_{+} - \Sigma_{-}) V^{\dagger} |\psi\rangle \end{pmatrix} = \begin{pmatrix} M_k(t) |\psi\rangle \\ |\phi\rangle \end{pmatrix}$$

The upper half of the output is exactly $M_k(t) |\psi\rangle$; $|\phi\rangle$ is a residue of the dilation and is discarded.

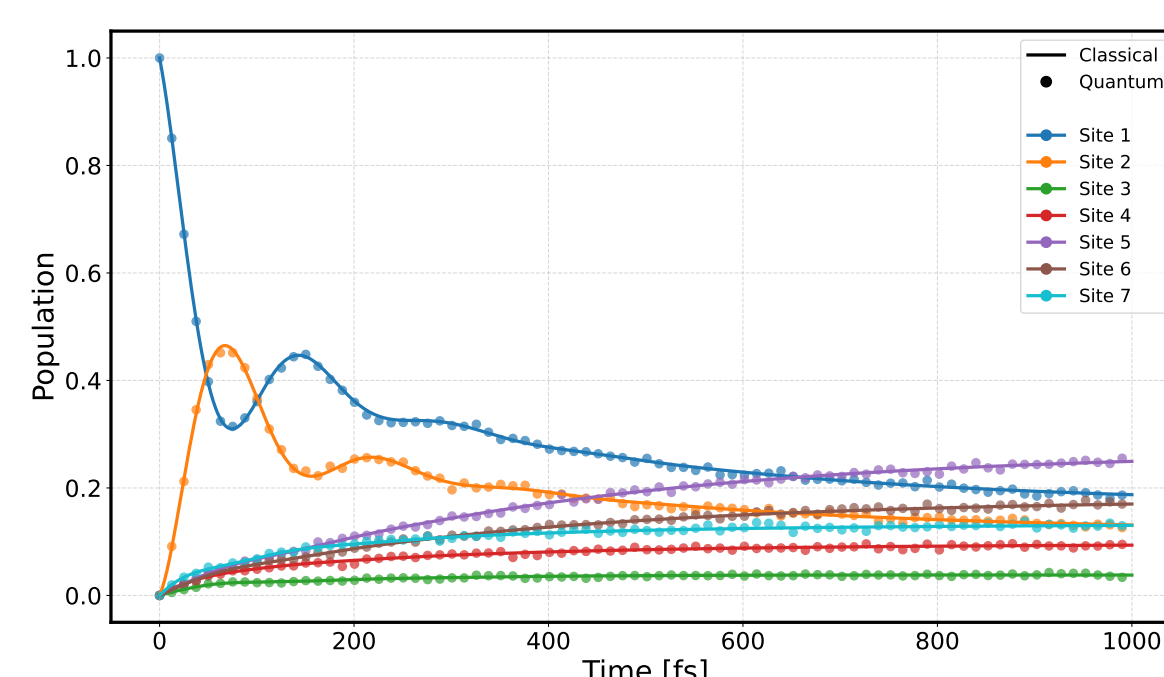


The loop: Lindblad $\rightarrow$ Choi $\rightarrow$ Kraus $\rightarrow$ SVD and dilation $\rightarrow$ circuit $\rightarrow \rho_S(t)$, then one step further in time.

The enhanced algorithm. In the loop above the state is re-prepared classically at every time step. Instead, the state of the *previous* step is reused:

- the **same** circuit is applied again and again
- the classical workload drops, the work moves to the QPU
- the growth of the classical cost with t disappears

Result: Markovian Dynamics



Site populations of the seven-site FMO complex over 1000 fs. Lines: classical reference, dots: quantum simulation.

Non-Markovian Dynamics: HEOM

The Markov approximation assumes a bath without memory. The **hierarchical equations of motion** drop it: the bath degrees of freedom are absorbed into **auxiliary density operators** (ADOs) $\hat{\sigma}_{\mathbf{n}}$ forming a coupled hierarchy,

$$\frac{d}{dt} \hat{\sigma}_{\mathbf{n}} = \underbrace{-\frac{i}{\hbar} \mathcal{L}_S \hat{\sigma}_{\mathbf{n}} + \sum_{j,k} (\phi_j \theta_{j,0} - \eta_{jk} \gamma_{jk}) \hat{\sigma}_{\mathbf{n}}}_{\text{own dynamics and decay}} + \underbrace{\sum_{j,k} \phi_j \hat{\sigma}_{\mathbf{n}_{jk}}}_{\text{level above}} + \underbrace{\sum_{j,k} \eta_{jk} \theta_{j,k} \hat{\sigma}_{\mathbf{n}_{jk}}}_{\text{level below}}$$

$\phi_j, \theta_{j,k}$: bath-coupling superoperators, γ_{jk} : decay rate of mode jk . Level 0 is ρ_S itself.

Numerically exact – no Born, Markov or secular approximation; the only error is the truncation depth $\mathcal{N}$:

$$\# \text{ ADOs} = \frac{(\mathcal{N} + NK)!}{\mathcal{N}! (NK)!}$$

N sites, K bath exponentials – grows fast, HEOM is expensive.

The Non-Markovian Propagator on a Circuit

Following Dan *et al.* [5]: HEOM supplies a propagator $G(t)$ whose m -th column holds the populations for an initial excitation on site m :

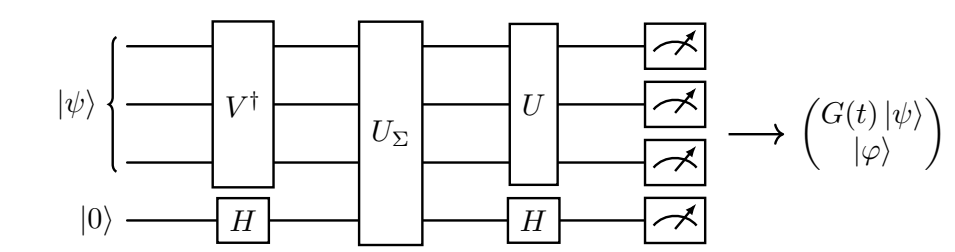
$$|\psi(t)\rangle = G(t) |\psi(0)\rangle$$

$$G(t) = \left\{ \begin{pmatrix} \rho_{11}^1(t_0) & \rho_{11}^2(t_0) & \rho_{11}^3(t_0) & \rho_{11}^4(t_0) \\ \rho_{22}^1(t_0) & \rho_{22}^2(t_0) & \rho_{22}^3(t_0) & \rho_{22}^4(t_0) \\ \rho_{33}^1(t_0) & \rho_{33}^2(t_0) & \rho_{33}^3(t_0) & \rho_{33}^4(t_0) \\ \rho_{44}^1(t_0) & \rho_{44}^2(t_0) & \rho_{44}^3(t_0) & \rho_{44}^4(t_0) \end{pmatrix}, \begin{pmatrix} \rho_{11}^1(t_1) & \rho_{11}^2(t_1) & \rho_{11}^3(t_1) & \rho_{11}^4(t_1) \\ \rho_{22}^1(t_1) & \rho_{22}^2(t_1) & \rho_{22}^3(t_1) & \rho_{22}^4(t_1) \\ \rho_{33}^1(t_1) & \rho_{33}^2(t_1) & \rho_{33}^3(t_1) & \rho_{33}^4(t_1) \\ \rho_{44}^1(t_1) & \rho_{44}^2(t_1) & \rho_{44}^3(t_1) & \rho_{44}^4(t_1) \end{pmatrix}, \dots \right\}$$

Four-site case. Superscript: site of the initial excitation, hence $G(t_0) = \mathbb{1}$; G is a *list*, one matrix per step.

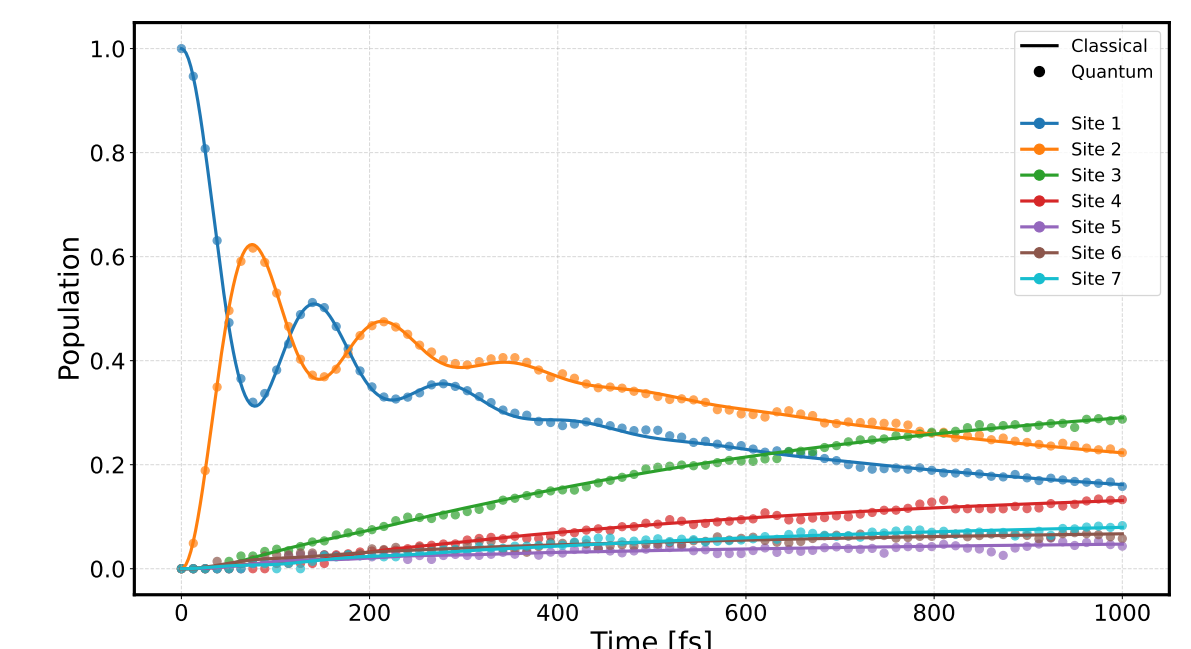
$$G(t) = U \Sigma V^{\dagger}$$

$G(t)$ is non-unitary $\Rightarrow$ the same SVD and the same dilation U_{Σ} as above.



$U, V^{\dagger}$ come straight from the SVD of $G(t)$; $G(t)$ itself is computed with HEOM.

$$H U_{\Sigma} V^{\dagger} H |\psi(0)\rangle \otimes |0\rangle = \begin{pmatrix} |\psi(t)\rangle \\ |\phi\rangle \end{pmatrix}, \quad \rho_S = |\psi(t)\rangle\langle\psi(t)|$$



Seven-site FMO, non-Markovian: the coherent oscillation between sites 1 and 2 survives far longer than in the Markovian case.

One Continuous Grid on Real Hardware

This work. Stack ρ_S and all ADOs into one super-vector $\vec{\Sigma} = (\text{vec } \rho_S, \text{vec } \sigma_1, \dots)^T$. There the evolution is exact and local in time – the ADOs carry the memory, so the propagator needs no memory kernel and P^n stays exact for every n .

Obstacle. A Sz.-Nagy dilation of P costs a success probability s^{-2n} with $s = \|P\|_2 = 32.39$: for 200 steps 10^{-301} – hopeless, though the dynamics itself stays bounded.

Four-site FMO on ibm_kingston, 70 circuits, $\Delta t = 20$ fs. Top: populations, bottom: coherence ρ_{12} ; lines classical, squares hardware.

$$\frac{d}{dt} \vec{\Sigma} = \mathcal{L}_{\text{HEOM}} \vec{\Sigma} \Rightarrow P = e^{\mathcal{L}_{\text{HEOM}} \Delta t}, \quad \vec{\Sigma}_n = P^n \vec{\Sigma}_0$$

P forms a **semigroup**: P^n is exact for every n .

Fix: a metric in which $\mathcal{L}$ is dissipative. $\|P\|_2 = 32$ says something about the Euclidean inner product, not about the physics – a Lyapunov solve gives a positive definite W :

$$\tilde{P} = W^{1/2} P W^{-1/2}, \quad \|\tilde{P}\|_2 \leq e^{\delta \Delta t} \approx 1$$

An exact similarity transform; success probability $\mathcal{O}(1)$.

Compression is now free: a Krylov projection of a contraction is again one, $\tilde{H}_m = \tilde{Q}_m^{\dagger} \tilde{P} \tilde{Q}_m$.

$$U = \begin{pmatrix} \tilde{H}_m/s & \sqrt{I - \tilde{H}_m \tilde{H}_m^{\dagger}/s^2} \\ \sqrt{I - \tilde{H}_m^{\dagger} \tilde{H}_m/s^2} & -(\tilde{H}_m/s)^{\dagger} \end{pmatrix}$$

$m = 4 \Rightarrow$ **3 qubits**, 31 CNOTs per step: **one single unitary**, re-applied n times. The classical cost is paid once, independent of the number of steps.

References

- [1] J. Adolphs, T. Renger, *How Proteins Trigger Excitation Energy Transfer in the FMO Complex of Green Sulfur Bacteria*, Biophys. J. **91**, 2778 (2006).
- [2] C. Kreisbeck, *Quantum Transport through Complex Networks*, PhD thesis, Universität Regensburg (2012).
- [3] D. M. Wilkins, *A Theoretical Investigation into Energy Transfer in Photosynthetic Open Quantum Systems*, University of Oxford (2012); arXiv:1503.03277.
- [4] A. Seneviratne, P. L. Walters, F. Wang, *Exact Non-Markovian Quantum Dynamics on the NISQ Device Using Kraus Operators*, ACS Omega **9**, 9666 (2024).
- [5] X. Dan, E. Geva, V. S. Batista, *Simulating Non-Markovian Quantum Dynamics on NISQ Computers Using the Hierarchical Equations of Motion*, J. Chem. Theory Comput. **21**, 1530 (2025).